Vikas Gurjar

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EDUCATION

Vellore Institute of Technology (VIT) — **2022 – Present**
B.Tech in Computer Science and Engineering — CGPA: 8.09 — *Madhya Pradesh, India*

TECHNICAL SKILLS

Languages: Python, Java, SQL
AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, NLTK
Generative AI: RAG (Retrieval-Augmented Generation), FAISS Vector DB, Google Gemini API, LangChain, Embeddings
Backend & APIs: FastAPI, Flask, RESTful Services, Asynchronous Programming
MLOps & Infrastructure: Docker, MLflow, GitHub Actions (CI/CD), Git, AWS (EC2)
Databases & Tools: MySQL, Redis, SQLite, Linux (Bash Scripting), Streamlit

PROJECTS

Enterprise AI Copilot – Private Document Intelligence Platform — **Sep 2025 – Dec 2025**
Python, FastAPI, FAISS, Gemini, Redis, Docker

- Engineered a **production-grade RAG system** to query 1,000+ enterprise documents, reducing information retrieval time by **60%**.
- Implemented **FAISS HNSW indexing** and semantic search, achieving a sub-200ms retrieval latency and a **Recall@10 of 92%**.
- Designed a modular backend with **FastAPI**, enabling persistent conversation memory via **Redis** for context-aware responses.
- Containerized the application using **Docker** and set up health monitoring, ensuring **99.9% uptime** during load testing.

Real-Time Financial Anomaly Detection Pipeline — **March 2025 – June 2025**
Python, Scikit-learn, MLflow, Docker, CI/CD

- Developed a high-throughput ML pipeline to detect fraudulent transactions, handling **500+ transactions per second (TPS)**.
- Solved data imbalance issues using **SMOTE** (Synthetic Minority Over-sampling Technique), improving the model's F1-score from 0.72 to **0.89**.
- Integrated **MLflow** for experiment tracking and model registry, reducing model versioning errors by **100%**.
- Automating testing and deployment via **GitHub Actions (CI/CD)**, reducing deployment cycle time by **40%**.

Autonomous Lane Detection System (Computer Vision) — **Oct 2024 – Jan 2025**
Python, OpenCV, NumPy

- Built a lane detection algorithm using **Canny Edge Detection** and **Hough Transform**, achieving **95% accuracy** on test video feeds.
- Optimized the image processing pipeline to run at **35 FPS (Real-time)** on standard CPU hardware by implementing ROI (Region of Interest) masking.
- Implemented adaptive thresholding to maintain detection consistency across varying lighting conditions (day/night/shadows).

EXTRACURRICULAR ACTIVITIES

- **BlockDAG Hackathon 2025 (Round 2 Qualifier):** Built **GhostWallet**, a decentralized privacy-focused wallet, integrating smart contracts for secure transactions.
- **Open Source Contribution:** Active contributor to developer communities; engaged in optimizing code repositories and documentation.
- **Volleyball Team Captain:** Represented college in inter-university tournaments; led team strategy and training sessions.

CERTIFICATIONS

- OCI AI Foundations — Oracle University (2025)
- Generative AI using IBM Watsonx — IBM Career Education Program (2025)
- Cyber Security Analyst — IBM Career Education Program (2025)